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Lab Practical #01:

Study of basic networking commands and IP configuration.

Practical Assignment #01:

1. Perform and explain various networking commands listed below:
   i. ipconfig
   ii. ping
   iii. getmac
   iv. systeminfo
   v. traceroute / tracert
   vi. netstat
   vii. nslookup
   viii. hostname
   ix. pathping
   x. arp

1. ipconfig

Description:

Ipconfig is a Windows command-line tool used to display and manage the computer's IP address, subnet mask, default gateway, and other TCP/IP configuration settings. It is helpful for diagnosing and troubleshooting network issues.

No.	Option	Description
1	ipconfig	Shows your IP address and basic network info.
2	ipconfig /all	Shows full details like DNS, MAC address, etc.
3	ipconfig /release	It is disconnects the internet from your IP.
4	ipconfig /renew	Gets a new IP address from the network.
5	ipconfig /flushdns	Fixes website loading issues by clearing DNS.

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Implementation:

```
C:\Windows\System32>ipconfig /all google.com

Error: unrecognized or incomplete command line.

USAGE:
    ipconfig [/allcompartments] [/? | /all |
                                 /renew [adapter] | /release [adapter] |
                                 /renew6 [adapter] | /release6 [adapter] |
                                 /flushdns | /displaydns | /registerdns |
                                 /showclassid adapter |
                                 /setclassid adapter [classid] |
                                 /showclassid6 adapter |
                                 /setclassid6 adapter [classid] ]

where
    adapter             Connection name
                       (wildcard characters * and ? allowed, see examples)

    Options:
       /?               Display this help message
       /all             Display full configuration information.
       /release         Release the IPv4 address for the specified adapter.
       /release6        Release the IPv6 address for the specified adapter.
       /renew           Renew the IPv4 address for the specified adapter.
       /renew6          Renew the IPv6 address for the specified adapter.
       /flushdns        Purges the DNS Resolver cache.
       /registerdns     Refreshes all DHCP leases and re-registers DNS names
       /displaydns      Display the contents of the DNS Resolver Cache.
       /showclassid     Displays all the dhcp class IDs allowed for adapter.
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```
C:\Windows\System32>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . . : 2409:40c1:3114:d9bd:5d76:b901:383f:3fc8
   Temporary IPv6 Address. . . . . . . : 2409:40c1:3114:d9bd:2d84:1775:db95:1ca9
   Link-local IPv6 Address . . . . . . : fe80::c3b4:f7f9:c27a:f1c6%19
   Default Gateway . . . . . . . . . . : fe80::e0a4:62ff:fe23:d446%19
```

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```
C:\Windows\System32>ipconfig /renew

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . . : 2409:40c1:3114:d9bd:5d76:b901:383f:3fc8
   Temporary IPv6 Address. . . . . . . : 2409:40c1:3114:d9bd:2d84:1775:db95:1ca9
   Link-local IPv6 Address . . . . . . : fe80::c3b4:f7f9:c27a:f1c6%19
   IPv4 Address. . . . . . . . . . . . : 192.168.126.231
   Subnet Mask . . . . . . . . . . . . : 255.255.255.0
```

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2. ping

Description:

The ping command is used to test the connection between your computer and another device or website.

No.	Option	Description
1	ping -a	Resolves the hostname from an IP address (shows the device name).
2	ping -t	Keeps pinging the target until you stop it manually.
3	ping -l	Sets the size of the ping packet. Example: ping -l 100 google.com
4	ping -n	Sets the number of ping attempts. Example: ping -n 5 google.com sends 5 pings
5	ping 127.0.0.1	It is used to test if your IP is working or not.

Implementation:

```
C:\Windows\System32>ping -a google.com

Pinging google.com [2404:6800:4009:823::200e] with 32 bytes of data:
Reply from 2404:6800:4009:823::200e: time=205ms
Reply from 2404:6800:4009:823::200e: time=92ms
Reply from 2404:6800:4009:823::200e: time=122ms
Reply from 2404:6800:4009:823::200e: time=178ms

Ping statistics for 2404:6800:4009:823::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 92ms, Maximum = 205ms, Average = 149ms
```

```
C:\Windows\System32>ping -n 3 google.com

Pinging google.com [2404:6800:4009:802::200e] with 32 bytes of data:
Reply from 2404:6800:4009:802::200e: time=435ms
Reply from 2404:6800:4009:802::200e: time=305ms
Reply from 2404:6800:4009:802::200e: time=247ms

Ping statistics for 2404:6800:4009:802::200e:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 247ms, Maximum = 435ms, Average = 329ms
```

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```
C:\Windows\System32>ping -l 4 google.com

Pinging google.com [2404:6800:4009:823::200e] with 4 bytes of data:
Reply from 2404:6800:4009:823::200e: time=132ms
Reply from 2404:6800:4009:823::200e: time=189ms
Reply from 2404:6800:4009:823::200e: time=180ms
Reply from 2404:6800:4009:823::200e: time=54ms

Ping statistics for 2404:6800:4009:823::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 54ms, Maximum = 189ms, Average = 138ms
```

```
C:\Windows\System32>ping -t google.com

Pinging google.com [2404:6800:4009:823::200e] with 32 bytes of data:
Reply from 2404:6800:4009:823::200e: time=151ms
Reply from 2404:6800:4009:823::200e: time=80ms
Reply from 2404:6800:4009:823::200e: time=276ms
Reply from 2404:6800:4009:823::200e: time=146ms
Reply from 2404:6800:4009:823::200e: time=91ms
Reply from 2404:6800:4009:823::200e: time=158ms
Reply from 2404:6800:4009:823::200e: time=64ms
Reply from 2404:6800:4009:823::200e: time=53ms
Reply from 2404:6800:4009:823::200e: time=87ms
Reply from 2404:6800:4009:823::200e: time=133ms

Ping statistics for 2404:6800:4009:823::200e:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 53ms, Maximum = 276ms, Average = 123ms
```

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3. getmac

Description:

The getmac command is used to display the MAC address (physical address) of your computer's network adapter. The MAC address is a unique ID used to identify your device on a network.

No.	Option	Description
1	getmac	Shows the MAC address of all network adapters on your computer.
2	getmac /v	Displays MAC address with extra details (like connection name and transport name).
3	getmac /fo table	Shows the output in a clean table format.
4	getmac /fo list	Shows the output in a list format.
5	getmac /fo csv	Displays the output in CSV format.

Implementation:

```
C:\Windows\System32>getmac /fo list

Physical Address: BC-0F-F3-78-0C-23
Transport Name:   Media disconnected

Physical Address: N/A
Transport Name:   Media disconnected

Physical Address: C8-5E-A9-2B-C8-25
Transport Name:   \Device\Tcpip_{C055FD37-FAD3-4FE3-A9F8-C337C458931D}
```

```
C:\Windows\System32>getmac /fo table

Physical Address    Transport Name
=================== ==========================================================
BC-0F-F3-78-0C-23   Media disconnected
N/A                 Media disconnected
C8-5E-A9-2B-C8-25   \Device\Tcpip_{C055FD37-FAD3-4FE3-A9F8-C337C458931D}
```

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```
C:\Windows\System32>getmac /v

Connection Name Network Adapter Physical Address    Transport Name
=============== =============== =================== ============================================================
Ethernet        Realtek Gaming  BC-0F-F3-78-0C-23   Media disconnected
Local Area Conn ExpressVPN TUN  N/A                 Media disconnected
Wi-Fi           Intel(R) Wi-Fi  C8-5E-A9-2B-C8-25   \Device\Tcpip_{C055FD37-FAD3-4FE3-A9F8-C337C458931D}
```

```
C:\Windows\System32>getmac

Physical Address    Transport Name
=================== ============================================================
BC-0F-F3-78-0C-23   Media disconnected
N/A                 Media disconnected
C8-5E-A9-2B-C8-25   \Device\Tcpip_{C055FD37-FAD3-4FE3-A9F8-C337C458931D}
```

4. systeminfo

Description:

The systeminfo command shows detailed information about your computer system, including the OS version, processor, memory, network, and more.

No.	Option	Description
1	systeminfo	Shows complete system details like OS, CPU, RAM, BIOS, etc.
2	systeminfo /s <hostname>	Gets system info of a remote computer. Replace <hostname> with name.
3	systeminfo /u <user>	Specifies the user to use when connecting to a remote system.
4	systeminfo /p <password>	Provides the password for the user in /u option.

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Implementation:

```
C:\Windows\System32>systeminfo

Host Name:                     LAPTOP-117LAL9V
OS Name:                       Microsoft Windows 11 Home Single Language
OS Version:                    10.0.26100 N/A Build 26100
OS Manufacturer:               Microsoft Corporation
OS Configuration:              Standalone Workstation
OS Build Type:                 Multiprocessor Free
Registered Owner:              pujan
Registered Organization:       HP
Product ID:                    00356-24687-84378-AAOEM
Original Install Date:         26-12-2024, 21:24:40
System Boot Time:              10-08-2025, 11:26:18
System Manufacturer:           HP
System Model:                  Victus by HP Gaming Laptop 15-fa0xxx
System Type:                   x64-based PC
Processor(s):                  1 Processor(s) Installed.
                               [01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2500 Mhz
BIOS Version:                  AMI F.28, 19-08-2024
Windows Directory:             C:\WINDOWS
System Directory:              C:\WINDOWS\system32
Boot Device:                   \Device\HarddiskVolume1
System Locale:                 en-us;English (United States)
Input Locale:                  00004009
Time Zone:                     (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:         7,860 MB
Available Physical Memory:     1,518 MB
Virtual Memory: Max Size:      18,612 MB
Virtual Memory: Available:     9,431 MB
Virtual Memory: In Use:        9,181 MB
Page File Location(s):         D:\pagefile.sys
Domain:                        WORKGROUP
Logon Server:                  \\LAPTOP-117LAL9V
Hotfix(s):                     3 Hotfix(s) Installed.
                               [01]: KB5056579
                               [02]: KB5062660
                               [03]: KB5064485
Network Card(s):               3 NIC(s) Installed.
                               [01]: Realtek Gaming GbE Family Controller
                                     Connection Name: Ethernet
                                     Status:          Media disconnected
                               [02]: ExpressVPN TUN Driver
                                     Connection Name: Local Area Connection
                                     Status:          Media disconnected
                               [03]: Intel(R) Wi-Fi 6E AX211 160MHz
                                     Connection Name: Wi-Fi
                                     DHCP Enabled:    Yes
                                     DHCP Server:     10.88.247.82
```

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```
C:\Windows\System32>systeminfo /?

SYSTEMINFO [/S system [/U username [/P [password]]]] [/FO format] [/NH]

Description:
    This tool displays operating system configuration information for
    a local or remote machine, including service pack levels.

Parameter List:
    /S      system          Specifies the remote system to connect to.

    /U      [domain\]user   Specifies the user context under which
                            the command should execute.

    /P      [password]      Specifies the password for the given
                            user context. Prompts for input if omitted.

    /FO     format          Specifies the format in which the output
                            is to be displayed.
                            Valid values: "TABLE", "LIST", "CSV".

    /NH                     Specifies that the "Column Header" should
                            not be displayed in the output.
                            Valid only for "TABLE" and "CSV" formats.

    /?                      Displays this help message.

Examples:
    SYSTEMINFO
    SYSTEMINFO /?
    SYSTEMINFO /S system
    SYSTEMINFO /S system /U user
    SYSTEMINFO /S system /U domain\user /P password /FO TABLE
    SYSTEMINFO /S system /FO LIST
    SYSTEMINFO /S system /FO CSV /NH
```

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5. traceroute / tracert

Description:

The tracert command is used to trace the path that data takes from your computer to a destination**.** It helps you find where delays or problems happen in the network.

No.	Option	Description
1	tracert -d	Doesn't resolve IP addresses to hostnames (faster results).
2	tracert -w	Sets the wait time (in milliseconds) for each reply. Example: -w 500 waits 0.5s.
3	tracert -h	Sets the maximum number of hops (network stops) to search. Default is 30.
4	tracert <hostname>	Traces the route to a specific website or IP. Example: tracert google.com
5	tracert -4	Forces tracert to use only IPv4 addresses.

Implementation:

```
C:\Windows\System32>tracert -d google.com

Tracing route to google.com [2404:6800:4009:82a::200e]
over a maximum of 30 hops:

  1   154 ms    48 ms    49 ms  2409:40c1:3146:953::3c
  2    78 ms   148 ms   120 ms  2405:200:5210:5:3924:110:3:403
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6   122 ms   103 ms    95 ms  2405:200:801:2e00::86
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9   118 ms   144 ms   150 ms  2001:4860:1:1::f48
 10   281 ms    71 ms    94 ms  2001:4860:1:1::f48
 11   374 ms   263 ms   121 ms  2404:6800:8129::1
 12     *        *       81 ms  2001:4860:0:1::53aa
 13   109 ms   200 ms   149 ms  2001:4860:0:1::77d4
 14   107 ms    90 ms   107 ms  2001:4860::9:4001:7734
 15   222 ms   197 ms   213 ms  2001:4860::9:4000:d773
 16   175 ms   242 ms   214 ms  2001:4860:0:1::6d1
 17   224 ms   203 ms   245 ms  2404:6800:4009:82a::200e

Trace complete.
```

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```
C:\Windows\System32>tracert -h 4 google.com

Tracing route to google.com [2404:6800:4009:80c::200e]
over a maximum of 4 hops:

  1     6 ms     3 ms     3 ms  2409:40c1:3146:953::3c
  2    28 ms    35 ms    38 ms  2405:200:5210:5:3924:110:3:403
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.

Trace complete.
```

```
C:\Windows\System32>tracert -w 4 google.com

Tracing route to google.com [2404:6800:4009:80c::200e]
over a maximum of 30 hops:

  1     6 ms     4 ms     3 ms  2409:40c1:3146:953::3c
  2    57 ms   139 ms     *     2405:200:5210:5:3924:110:3:403
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    51 ms    37 ms    28 ms  2405:200:801:2e00::84
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9    87 ms    84 ms    80 ms  2001:4860:1:1::2218
 10    93 ms    90 ms     *     2001:4860:1:1::2218
 11    79 ms    75 ms    81 ms  2404:6800:812b::1
 12     *       91 ms    85 ms  2001:4860:0:1::53a2
 13    88 ms    75 ms    76 ms  2001:4860:0:1::78bc
 14    95 ms    82 ms    88 ms  2001:4860::9:4001:7734
 15     *      104 ms   132 ms  2001:4860::9:4002:d931
 16     *        *        *     Request timed out.
 17   150 ms   163 ms    96 ms  2001:4860:0:1::4fe5
 18     *       87 ms    90 ms  bom07s18-in-x0e.1e100.net [2404:6800:4009:80c::200e]
```

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6. netstat

Description:

netstat (**Network Statistics**) is a command-line tool that shows network connections, open ports, routing tables, and network statistics. It helps in monitoring and troubleshooting network-related issues.

No.	Option	Description
1	netstat -a	Displays all active connections and listening ports.
2	netstat -b	Shows the executable program using each connection but requires admin rights
3	netstat -c	it refreshes output continuously.
4	netstat -n	Shows IP addresses and port numbers without resolving hostnames.
5	netstat -d	Shows statistics for each network interface (like Ethernet or Wi-Fi). Useful for checking packet errors, drops, and traffic load.

Implementation:

```
C:\Windows\System32>netstat -d

Active Connections

  Proto  Local Address          Foreign Address        State        DSCP Value
TCP    192.168.172.231:49408  4.213.25.241:https     ESTABLISHED        0
TCP    127.0.0.1:49686        LAPTOP-117LAL9V:49687  ESTABLISHED        0
TCP    127.0.0.1:49687        LAPTOP-117LAL9V:49686  ESTABLISHED        0
TCP    127.0.0.1:49688        LAPTOP-117LAL9V:49689  ESTABLISHED        0
TCP    127.0.0.1:49689        LAPTOP-117LAL9V:49688  ESTABLISHED        0
TCP    192.168.172.231:49716  ec2-54-169-5-36:http   ESTABLISHED        0
TCP    192.168.172.231:49717  ec2-54-169-7-73:http   ESTABLISHED        0
TCP    192.168.172.231:58231  bom07s27-in-f14:https  TIME_WAIT          0
TCP    192.168.172.231:58262  192.168.172.61:domain  FIN_WAIT_2         0
TCP    192.168.172.231:58263  192.168.172.61:domain  FIN_WAIT_2         0
TCP    192.168.172.231:58264  192.168.172.61:domain  FIN_WAIT_2         0
TCP    192.168.172.231:58337  ec2-54-255-172-2:http  TIME_WAIT          0
```

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7. nshookup

Description:

nslookup is a tool used to query DNS servers to find the IP address of a domain name or find the domain name of an IP address. It is useful for troubleshooting DNS problems.

No.	Option	Description
1	nslookup	Starts interactive mode (you can enter queries one by one).
2	nslookup google.com	Finds the IP address of google.com.
3	nslookup 8.8.8.8	Reverse lookup: Finds the domain name for IP address 8.8.8.8
4	nslookup -type	Shows server records of any website.
5	nslookup google.com 8.8.8.8	Queries google.com using the DNS server 8.8.8.8.

Implementation:

```
C:\Windows\System32>nslookup google.com
Server:  UnKnown
Address:  192.168.126.252

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4009:80b::200e
          142.250.70.78
```

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8. hostname

Description:

The hostname command is used to display or set the name of the computer (host) on a network

No.	Option	Description
1	hostname	Displays the current hostname for your computer.
2	hostname new-name	Temporarily change the hostname

Implementation:

```
C:\Windows\System32>hostname
LAPTOP-117LAL9V
```

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9. pathping

Description:

pathping is a network diagnostic command in Windows that combines the features of ping and tracert. It is useful for identifying network bottlenecks or connection issues.

No.	Option	Description
1	pathping -h	Sets the maximum number of hops to search.
2	pathping -g	Specifies a list of gateways to use for the test route.
3	pathping -i	Uses a specified source IP address.
4	pathping -n	Stops hostnames from being resolved (shows only Ips).
5	pathping -w	Sets wait time (in milliseconds) for each reply.

Implementation:

```
C:\Windows\System32>pathping -h 2 google.com

Tracing route to google.com [2404:6800:4009:82a::200e]
over a maximum of 2 hops:
  0  LAPTOP-117LAL9V [2409:40c1:3146:953:85c8:d244:41fd:9da3]
  1  2409:40c1:3146:953::3c
  2  2405:200:5210:5:3924:110:3:403

Computing statistics for 50 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           LAPTOP-117LAL9V [2409:40c1:3146:953:85c8:d244:41fd:9da3]
                                 0/ 100 =  0%   |
  1   14ms     0/ 100 =  0%     0/ 100 =  0%  2409:40c1:3146:953::3c
                                 0/ 100 =  0%   |
  2   42ms     0/ 100 =  0%     0/ 100 =  0%  2405:200:5210:5:3924:110:3:403
```

```
C:\Windows\System32>pathping -g google.com

Tracing route to google.com [142.250.70.78]
over a maximum of 30 hops:
  0  LAPTOP-117LAL9V [192.168.172.231]
  1     *        *        *
Computing statistics for 0 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           LAPTOP-117LAL9V [192.168.172.231]
```

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10.arp

Description:

The arp (Address Resolution Protocol) command is used to view and manage the ARP cache, which maps IP addresses to MAC addresses**.** It's helpful for diagnosing local network issues**.**

No.	Option	Description
1	arp -a	Shows the ARP table (IP-to-MAC address mappings) for all interfaces.
2	arp -g	Same as -a,displays the ARP entries.
3	arp -d	Deletes a specific ARP entry from the cache. Example: arp -d 192.168.1.1
4	arp -s	Adds a static ARP entry manually. Example: arp -s 192.168.1.5 00-AA-11-BB-22-CC
5	arp -v	Displays detailed output of the ARP table.

Implementation:

```
C:\Windows\System32>Arp -a

Interface: 192.168.172.231 --- 0x13
  Internet Address      Physical Address      Type
  192.168.172.61        8e-bc-ff-ba-bb-8d     dynamic
  192.168.172.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```